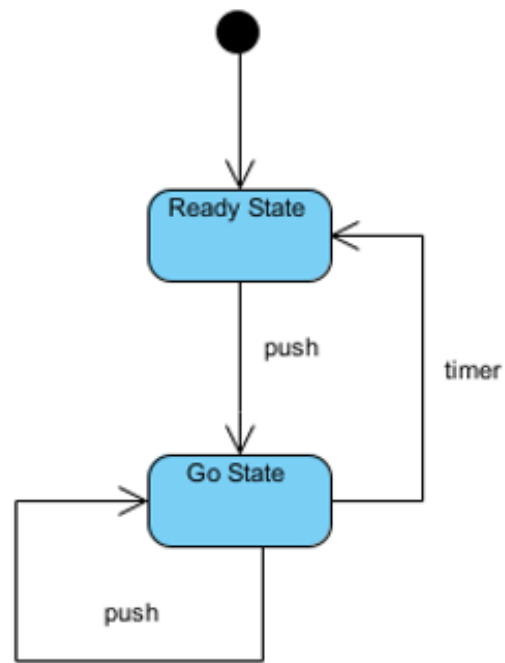


Lab 5

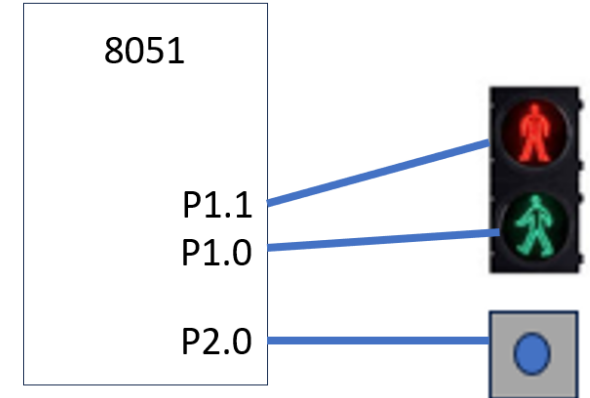
Coding State Machine Diagram

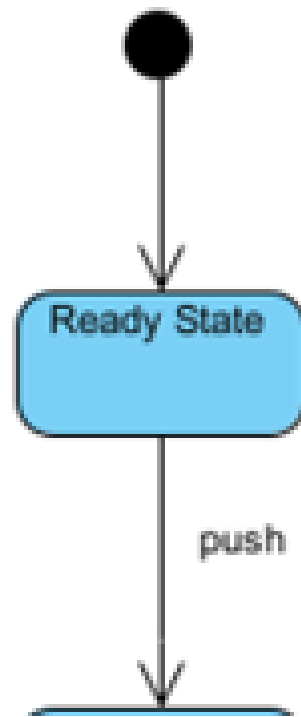
Traffic 1



Ready State

Go State





ReadyState:

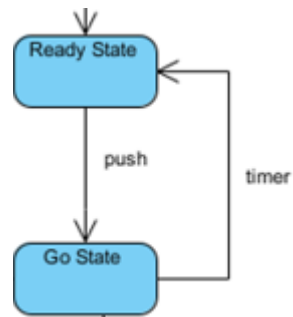
```
mov r0,#01H ;r0 =state 1
```

```
setb P1.1
```

```
clr p1.0
```

```
jB p2.0,GoState
```

```
sjmp ReadyState
```



GoState:

```
mov r0,#02H ;r0 =state 1
```

```
clr P1.1
```

```
setb p1.0
```

```
mov r1,#10
```

```
timer:
```

```
call delay
```

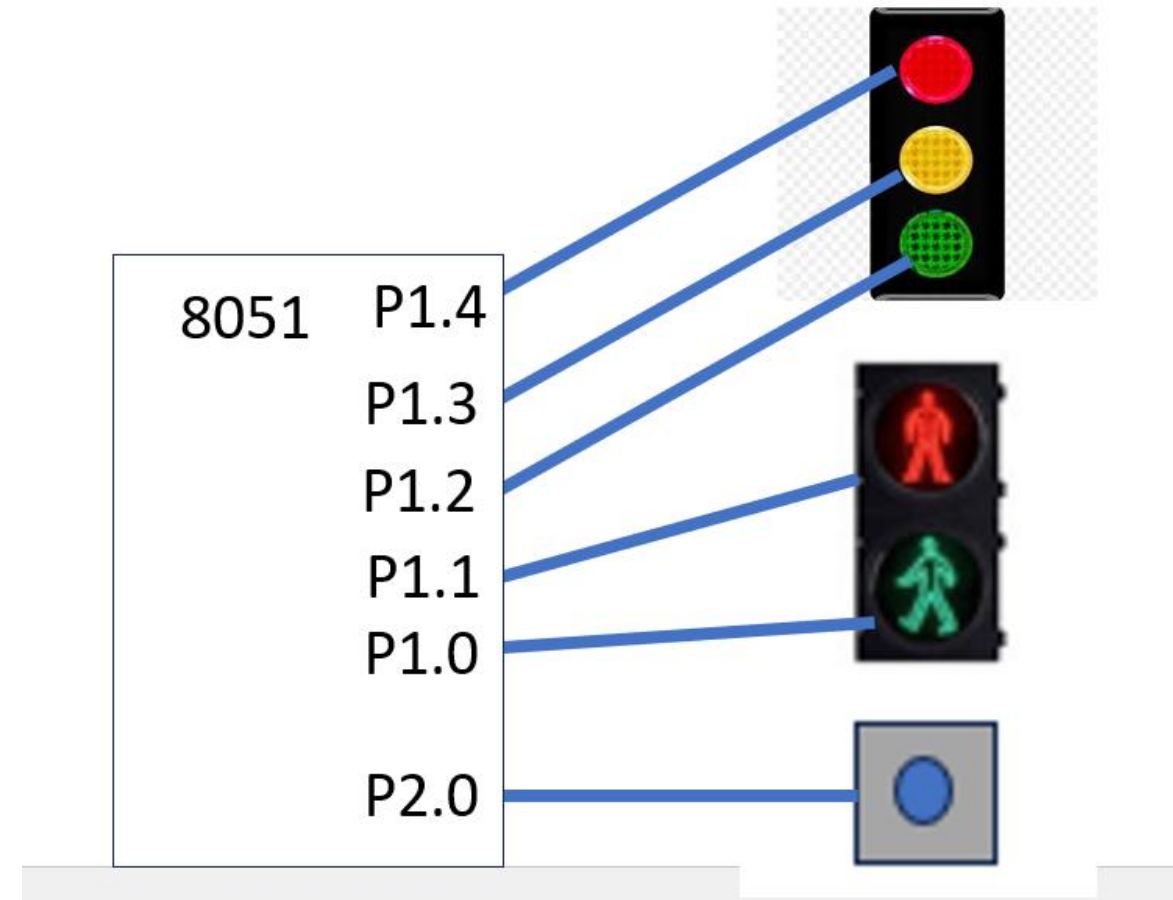
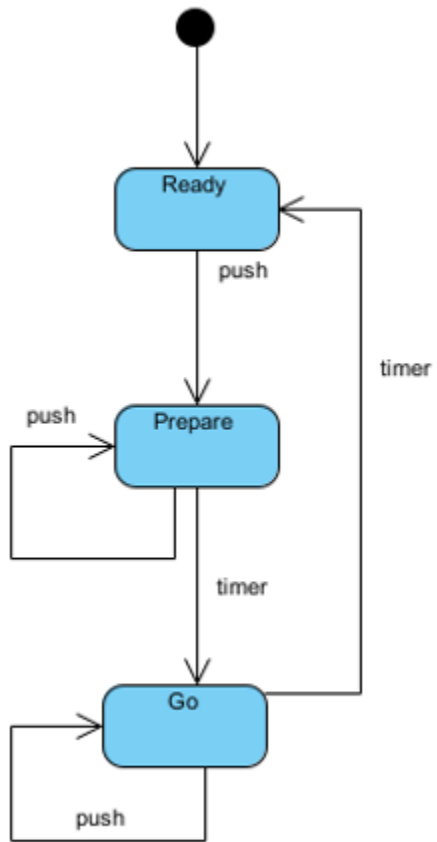
```
djnz r1,timer
```

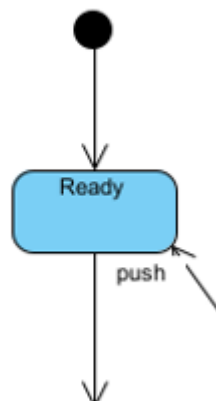
```
jB p2.0,GoState
```

```
sjmp Readystate
```

```
delay:
mov tmod,#01H ; timer0 model
mov th0,#11111111b;FFH
mov tl0,#11110110b;F6H delay 10usec
mov tcon,#10H;start timer
wait: jnb tcon.5,wait
mov tcon,#00H;stop timer
ret
```

Traffic 2





Ready State



ReadyState:

```
mov r0,#01H ;r0 =state 1
```

```
clr p1.4
```

```
clr P1.3
```

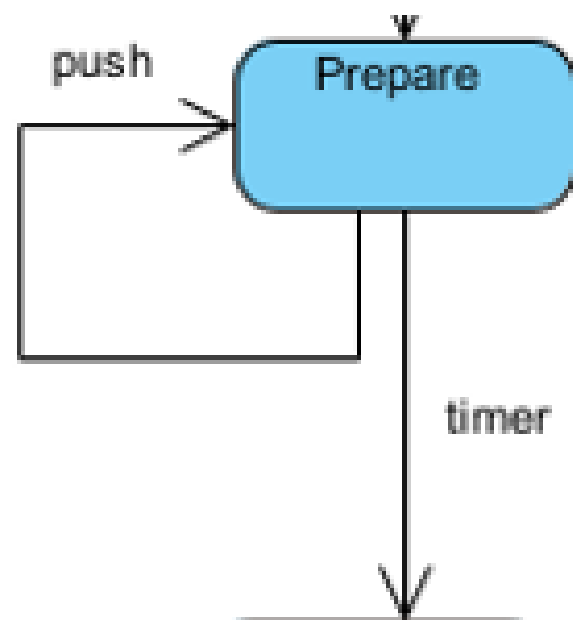
```
setb p1.2
```

```
setb P1.1
```

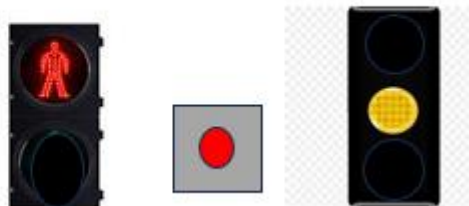
```
clr p1.0
```

```
jB p2.0,PrepareState
```

```
sjmp ReadyState
```



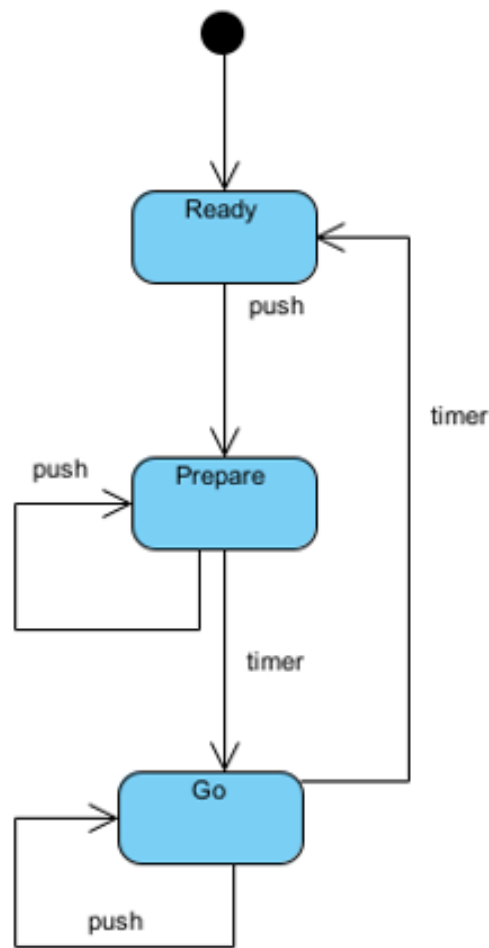
Prepare State



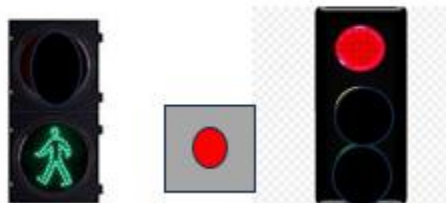
PrepareState:

```

mov r0,#02H ;r0 =state 2
clr p1.4
setb P1.3
clr p1.2
setb P1.1
clr p1.0
mov r1,#02 ; delay 20 usec
timerP:
call delay
djnz r1,timerP
jB p2.0,GoState
sjmp GoState
  
```



Go State



GoState:

```
mov r0,#03H ;r0 =state 3
```

```
setb p1.4
```

```
clr P1.3
```

```
clr p1.2
```

```
clr P1.1
```

```
setb p1.0
```

```
mov r1,#10 ; deley 100 usec
```

timer:

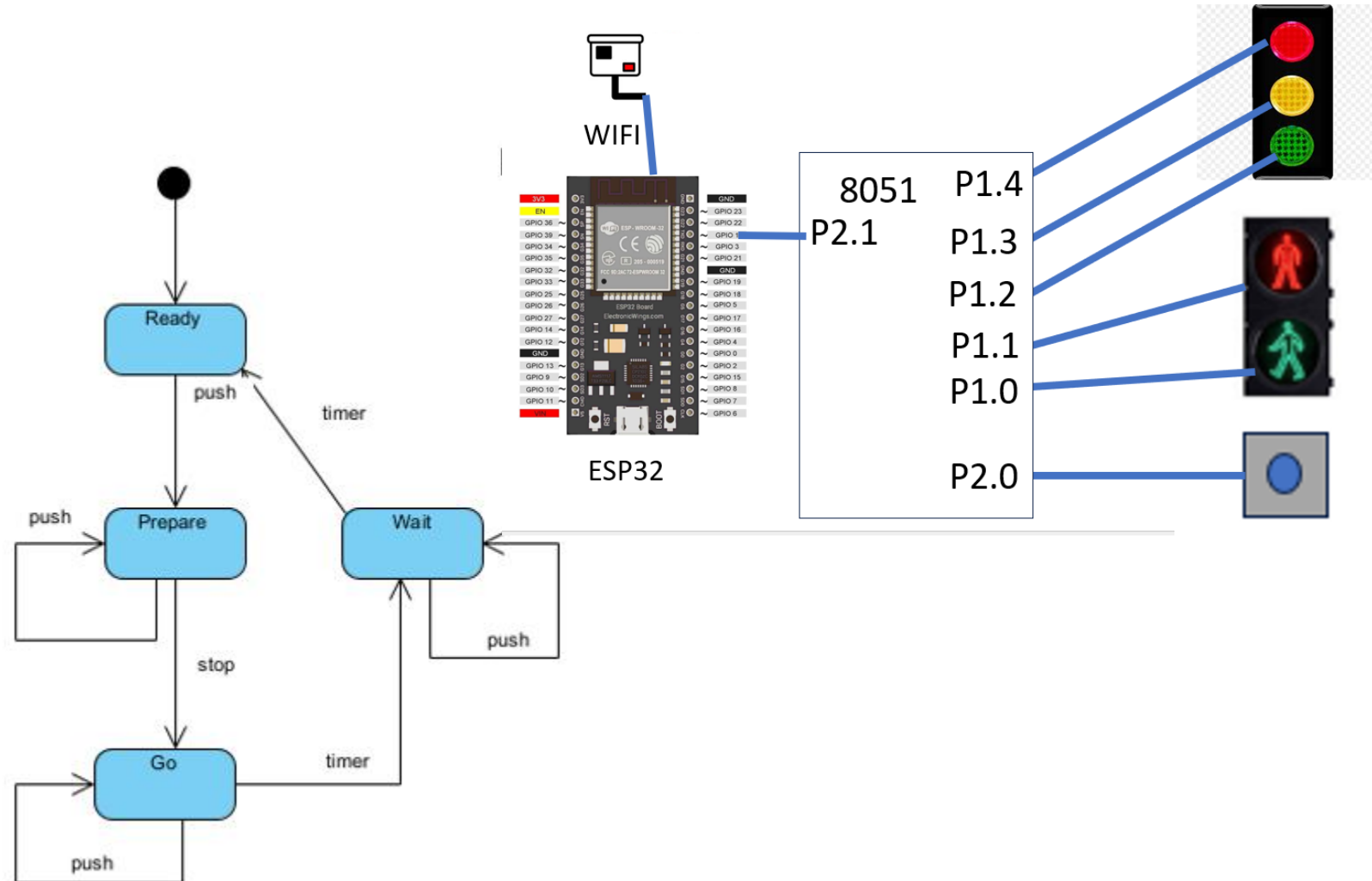
```
call delay
```

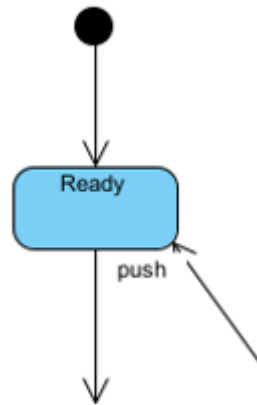
```
djnz r1,timer
```

```
jB p2.0,GoState
```

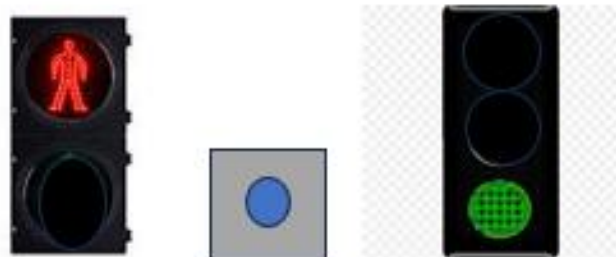
```
sjmp Readystate
```

Traffic 3





Ready State



ReadyState:

```
mov r0,#01H ;r0 =state 1
```

```
clr p1.4
```

```
clr P1.3
```

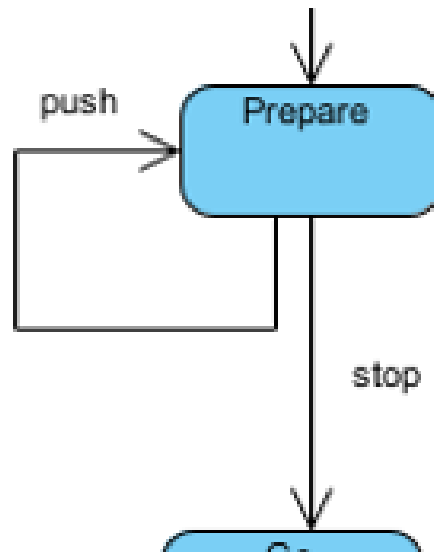
```
setb p1.2
```

```
setb P1.1
```

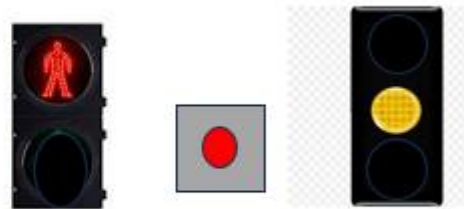
```
clr p1.0
```

```
jB p2.0,PrepareState
```

```
sjmp ReadyState
```



Prepare State



PrepareState:

```
mov r0, #02H ; r0 = state 2
```

```
clr p1.4
```

```
setb P1.3
```

```
clr p1.2
```

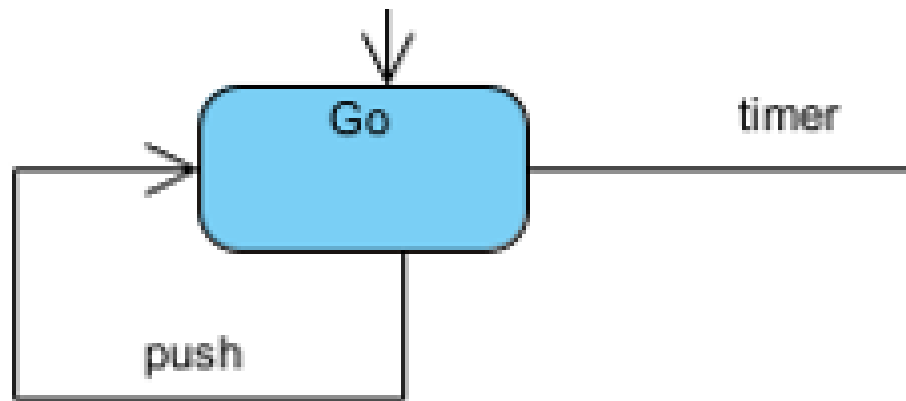
```
setb P1.1
```

```
clr p1.0
```

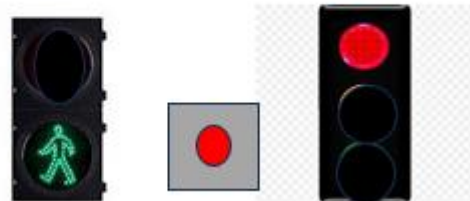
```
jB p2.1, GoState ; wait stop signal
```

```
sjmp PrepareState
```

```
|
```



Go State



GoState:

```
mov r0,#03H ;r0 =state 3
```

```
setb p1.4
```

```
clr P1.3
```

```
clr p1.2
```

```
clr P1.1
```

```
setb p1.0
```

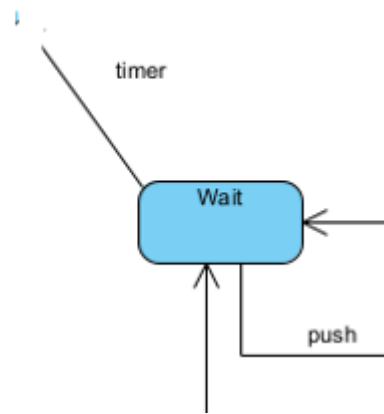
```
mov r1,#05 ; deley 50 usec
```

```
timer:
```

```
call delay
```

```
djnz r1,timer
```

```
sjmp WaitState
```



Wait State



WaitState:

```
mov r0,#04H ;r0 =state 4
```

```
clr p1.4
```

```
clr P1.3
```

```
setb p1.2
```

```
setb P1.1
```

```
clr p1.0
```

```
mov r1,#05 ; deley 50 usec
```

```
timerw:
```

```
call delay
```

```
djnz r1,timerw
```

```
sjmp readystate
```